heavy ion it may also become excited. However, the collision of the projectile and target forming the products does result in a value of Q different than zero. For light projectiles the process can be written:

$$X_1 + X_2 \left\langle \begin{array}{ll} a & \rightarrow X_1 + X_2 + \gamma \\ & \qquad Q < 0 \\ b & \rightarrow X_1 + X_2^* \end{array} \right. \qquad (10.24a) \quad (10.24b)$$

The reaction path *a* indicates that the energy Q is emitted as a γ-ray. In reaction path *b* the Q is retained as excitation energy of the target nuclide. The latter exists in an excited state and may transform to the ground state quite rapidly or may exist for a measurable time as an isomer. An example of inelastic scattering reaction (10.24b) is the formation of an isomer of silver by the irradiation of ^{107}Ag with neutrons

$$^{107}\mathrm{Ag(n, n')}^{107m}\mathrm{Ag} \rightarrow {}^{107}\mathrm{Ag} + \gamma \qquad (10.25)$$

the half-life of ^{107m}Ag is 44 s. The energy relationships in inelastic scattering are the same as for other nuclear reactions.

10.7. DISSECTING A NUCLEAR REACTION

As an example of a nuclear transmutation reaction, let us consider the following:

$$^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \rightarrow [{}^{18}_{9}\mathrm{F}]^* \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H} \qquad (10.26)$$

For pedagogic reasons we treat this reaction as if an intermediate *compound nucleus* is formed. The compound nucleus $^{18}_{9}$F is in square brackets to indicate its transitory nature and marked with an asterisk to indicate that it is excited. Induced nuclear reactions are often written in an abbreviated manner indicating first the target and then, in parentheses, the projectile and the smaller product, followed by the major product outside the parentheses. In the case of the sample reaction, we would write ^{14}N(α,p)^{17}O The abbreviations used for ^{4}He, ^{1}H, ^{2}H (=D), ^{3}H (=T), etc., are α, p, d, t, etc. The reactions may be classified by the particles in parentheses; the sample reaction is called an (α,p) type.

The sample reaction is of historical interest since this is actually the reaction studied by Rutherford in 1919 when he produced the first induced nuclear transformation in the laboratory. A cloud chamber photograph of the reaction is shown in Figure 10.6.

From the atomic rest masses we can calculate that the change in mass for this reaction is

$$\begin{aligned} \Delta M^0 &= M_3^0 + M_4^0 - M_1^0 - M_2^0 \\ &= (16.999\,131 + 1.007\,825 - 14.003\,074 - 4.002\,603) = 0.001\,279(\mathrm{u}) \end{aligned} \qquad (10.27)$$

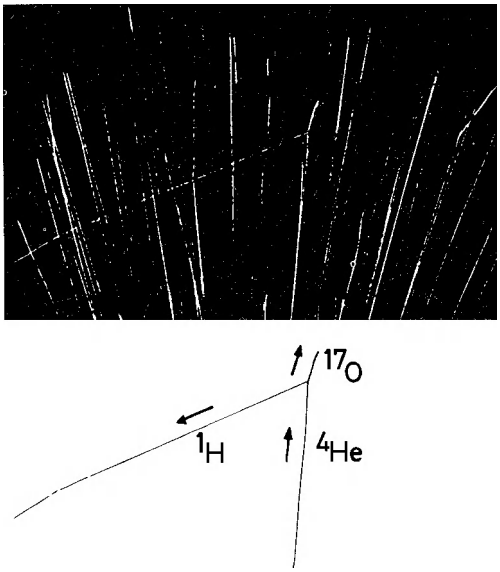


Figure 10.6 Cloud chamber photograph of the Rutherford reaction $^{4}He + {}^{14}N \rightarrow {}^{17}O + {}^{1}H$. *(From Blackett and Lees.)*

This increase in mass corresponds to a Q-value of -1.19 MeV (10.3). The energy required for this endoergic reaction can only be obtained through the kinetic energy of the projectile.

In the collision between the projectile and the target nucleus, which we have assumed to be stationary, the compound nucleus formed always acquires a certain kinetic energy, a recoil. The recoil energy can be calculated from the conservation laws to be

$$E_{\text{kin,c}} = E_{\text{kin,1}} M_1/(M_1 + M_2) \approx E_{\text{kin,1}} A_1/A_{\text{c}} \tag{10.28}$$

The mass numbers can be substituted for the atomic masses since we are carrying the calculation of the energy to three significant figures only. In order for the reaction to occur, the kinetic energy of the projectile $E_{\text{kin,1}}$ must exceed the kinetic energy of the compound nucleus $E_{\text{kin,c}}$ by the value of Q:

$$E_{\text{kin,1}} = E_{\text{kin,c}} - Q \tag{10.29}$$

The minimum projectile energy necessary for a reaction to occur is called the *threshold energy* E_{tr}, which according to (10.28) and (10.29) is

$$E_{\text{tr}} = -Q(M_1 + M_2)/M_2 \approx -QA_{\text{c}}/A_2 \quad (\text{if } Q < 0) \tag{10.30}$$

For our sample reaction (10.26), the threshold energy is $1.19 \times 18/14 = 1.53$ MeV.

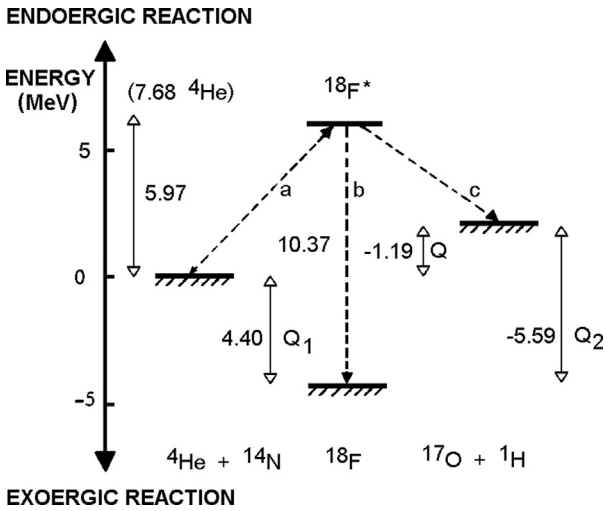


Figure 10.7 Energy diagram of the transmutation $^{4}He + {}^{14}N \rightarrow {}^{18}F^{*} \rightarrow {}^{17}O + {}^{1}H$ caused by bombarding ^{14}N with 7.68 MeV α-particles from ^{214}Po.

If the projectile has a higher kinetic energy than the threshold energy, the products would have a correspondingly higher combined kinetic energy, assuming no transformation into other forms of energy. Rutherford used α-particles from the radioactive decay of ^{214}Po, which have kinetic energies of 7.68 MeV. The products, therefore, had considerable kinetic energy, which is shown by the thick tracks in the cloud chamber picture of Figure 10.6.

In Figure 10.7 the energy pattern for our reaction is summarized. The Q-value for the first step of the reaction ($^{4}He + {}^{14}N \rightarrow {}^{18}F$) is $+4.40$ MeV (Q_1). The kinetic energy of the ^{4}He is 7.68 MeV, so conservation of momentum requires that the ^{18}F nucleus have $7.68 \times 4/(4 + 14) = 1.71$ MeV of kinetic energy. Thus, the internal excitation energy of $[{}^{18}F]^{*}$ is $4.40 + 7.68 - 1.71 = 10.37$ MeV. For the second step, $^{18}F \rightarrow {}^{17}O + {}^{1}H$, the Q_2-value is -5.59 MeV. This shows that the total excitation and kinetic energy of the products is $10.37 + 1.71 - 5.59 = 6.49$ MeV.

10.8. THE COMPOUND NUCLEUS MODEL

We have used a model which assumes the formation of an excited intermediary nucleus in the previous section to discuss some aspects of nuclear reactions. Let us now study this model a little closer.

If $E^{0}_{\text{kin}} > E_{\text{cb}}$ the attractive nuclear force dominates and the particle is absorbed by the target nucleus. Assuming $Q > 0$, the E^{0}_{mass} decreases. This means that E_{exc} increases, and the system is transformed into an excited compound nucleus. Moreover, since the projectile had a certain kinetic energy, the excitation of the compound nucleus is

$$E_{\text{exc}} = Q + E^{0}_{\text{kin}} \tag{10.29}$$